Validate Before You Estimate

Handout: three challenges and one question

Challenge 1: be the coder (3 minutes)

Code each bill title into exactly one topic using the codebook below. Choose the topic that best describes the main purpose of the bill; when a bill touches several topics, code its primary target. Then compare with your neighbor and circle the titles where you disagree.

Topic	Codebook (condensed from the Comparative Agendas Project)
Health	health care reform; insurance, Medicare and Medicaid; drug industry and approval; hospitals and clinics; health workforce; disease prevention and public health; mental health; long-term care; tobacco; treatment and prevention of drug and alcohol abuse; medical research
Education	higher education and student aid; elementary and secondary schools; underprivileged, vocational and special education; standards; education research
Environment	drinking water; waste and hazardous waste; air pollution and climate; recycling; species and forests; land and water conservation. Not energy, farming or the management of parks and public lands
Immigration	immigration, naturalization and citizenship; refugees and asylum; visas; enforcement of immigration law
Law and Crime	law enforcement agencies; white collar crime; drug trafficking and enforcement; courts; prisons and sentencing; juvenile crime; child abuse; family law and domestic violence; criminal and civil code; police
Social Welfare	low-income assistance including food assistance; the elderly including Social Security retirement; people with disabilities; charities and volunteers; child care
Other	everything else: economy and taxes, civil rights, agriculture, labor, energy, transportation, housing, commerce, defense and veterans, science, trade, foreign affairs, government operations, commemorations, public lands, culture, relief of named individuals

#	Bill title	Your code	Neighbor's code
1	(2009) To amend the Controlled Substances Act to provide for disposal of controlled substances by ultimate users and care takers through State take-back disposal programs, to amend the Federal Food, Drug, and Cosmetic Act to prohibit recommendations on drug labels for disposal by flushing, and for other purposes.		
2	(1976) A bill to supplement retirement benefits for State and local law enforcement officers and firefighters.		
3	(1976) A bill to amend the Internal Revenue Code of 1954 to provide that employed individuals over age 65 not be required to contribute to the Federal Insurance Contributions Act.		
4	(2015) A bill to amend the Internal Revenue Code of 1986 to modify and permanently extend qualified zone academy bonds, and to treat such bonds as specified tax credit bonds.		
5	(2010) To provide for certain enhanced border security measures, and for other purposes.		
6	(2004) To amend the Omnibus Parks and Public Lands Management Act of 1996 to extend the authorization for certain national heritage areas, and for other purposes.		

Bell question: on which title did you and your neighbor disagree, and what rule would have settled it?

Challenge 2, part one: place your bet

The same 1,650 titles were coded three times by the same cheap model with three prompts. Which one agrees most with the human coders? Tick one, then write a number.

	Instrument	Your guess of its accuracy
[ ]	P1 Category names only (no definitions)	____ %
[ ]	P2 The condensed codebook	____ %
[ ]	P3 The codebook plus 14 human-coded examples	____ %

Second bet: a stronger model with P3 beats the best cheap prompt by [ ] less than 5 points [ ] more than 5 points. Third bet: a lazy coder who answers "Other" every time scores ____ % accuracy.

Challenge 2, part two: whose error is it?

For each title the model and the human coder disagree. Vote, then say what you would change (a prompt rule, a codebook clarification, or the human code).

	Title	Human	Model	Who is wrong?
A	(2001) A bill to make an urgent supplemental appropriation for fiscal year 2001 for the Department of Defense for the Defense Health Program.	Health	Other	[ ] model [ ] human [ ] codebook
B	(1989) To implement the recommendations of the Interagency Committee and the Technical Study Group on Cigarette and Little Cigar Fire Safety, and for other purpose.	Other	Health	[ ] model [ ] human [ ] codebook
C	(2003) To amend title II of the Social Security Act to allow the Commissioner of Social Security reasonable discretion in applying the 10-year marriage requirement for entitlement of divorced spouses to spousal benefits in cases in which the divorce is in whole or in part the result of severe spousal or child abuse.	Law and Crime	Social Welfare	[ ] model [ ] human [ ] codebook

Challenge 3: how many titles would you code by hand? (exit ticket)

Your project: the same instrument on 20,000 bill titles. Your committee wants to know whether Democrats and Republicans differ by 3 points or more in the share of health bills. A research assistant codes about 100 titles an hour. Rule: to tell a 3-point gap between two groups from zero, each group's interval must be about plus or minus 2 points or narrower. The table is the curve on the screen (this lesson's error rates, projected to a group of 10,000 titles). It applies to each group separately.

Hand-coded titles in the group (n)	50	100	200	300	400	800
Corrected interval, plus or minus	5.2%	3.7%	2.7%	2.2%	1.9%	1.4%
Hand-coded sample alone, plus or minus	8.2%	5.8%	4.1%	3.4%	2.9%	2.1%

I would hand-code _____ titles per party, _____ titles in all, which is _____ RA hours. I would promise my committee an interval of about plus or minus _____ points per party. Without the model I would need _____ per party.

Question (part two): would you rather lose pairs or invent them?

Linking two files of people with no shared ID. The research question is the share of people whose postcode changed. Mark your choice and the reason.

Policy for the ambiguous pairs	Precision	Recall	Estimated share who moved	Your choice
Drop them (strict)	100.0%	86.0%	12.2%	[ ]
Keep them all (lenient)	84.6%	97.8%	22.3%	[ ]
Let a model adjudicate	99.5%	94.3%	14.2%	[ ]
Truth (every true link known)			15.6%	

Precision: when the rule says same person, how often it is right. Recall: of the true links, how many it finds. Materials, data and the notebook: Felipe H. Rodriguez Villalta, felipe.hrv@gmail.com